

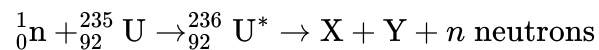
Chapter 30 Nuclear Energy and Elementary Particles

Process of Nuclear Energy

- Fission
Large mass number nucleus **splits** into two smaller nuclei
- Fusion
Two light nuclei **fuse** to form a heavier nucleus

Nuclear Fission

A heavy nucleus **splits** into two smaller nuclei



mid element is intermediate. X and Y are called fission fragments.

The fission fragments have less mass than original nuclei. This decrease in mass appears as released energy.

This fission releases much larger energy than chemical processes.

Chain Reaction

The neutrons emitted by fission **triggers fission of other nucleus**. This process is called a **chain reaction**.

If uncontrolled, a violent explosion.

If controlled, nuclear reactor

Nuclear Reactor

A system designed to maintain a **self-sustained** chain reaction.

The reproduction constant k is defined as the average number of neutrons from each fission that cause another fission ($k = 1$, critical; $k < 1$, subcritical; $k > 1$, supercritical)

Design

- Fuel elements: Enriched uranium
- Moderator: slow down the neutrons
- Control rods: absorb neutrons

Pressurized Water Reactor

Maintain water in high pressure to keep it from boiling.

- hot water in primary system pumped through a heat exchanger
- Heat transferred to the water in secondary system
- water converted into steam
- Steam create electric power
- water in secondary system isolated from in primary system to prevent pollution

Reactor Safety

- Three level of containment: vessel, building, location
- Emergency cooling system
- Disposal of waste material
- Transportation of fuel and waste

Nuclear Fusion

Two light nuclei combine to form a heavier nucleus.

The mass of final nucleus is less than the masses of the original nuclei.

Fusion in the Sun

Conditions

- High temperature
- High rate of collisions

Proton-Proton Cycle

A series of three nuclear reactions believed to operate in the Sun. Energy liberated is gamma rays, positrons and neutrinos.

Fusion Reactors

Advantages

- Inexpensive fuel source
- few by-products

Considerations:

- High temperature
- Overcome Coulomb repulsion

Requirements for Successful Thermonuclear Reactor

Plasma ion density n ; Plasma confinement time τ

Lawson's Criteria

- $n\tau \geq 10^{14} s/cm^3$ for deuterium-tritium
- $n\tau > 10^{16} s/cm^3$ for deuterium-deuterium

Methods of Creating Fusion Events

- Magnetic confinement
- Inertial laser confinement
- Inertial electrostatic confinement

Some Fusion Reactors

- TFTR
- JET
- NSTX
- ITER